

AMAN JAISWAL

AI/ML | Computer Vision | Healthcare AI | NLP | Multimodal AI

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[LinkedIn](#) | [Github](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building machine learning systems across healthcare AI, computer vision, multilingual NLP, and generative AI. Experienced in deep learning pipelines, FastAPI deployment, signal processing, and scalable AI experimentation. Research exposure at IIT Ropar with strong performance in national-level hackathons and real-world AI development.

EDUCATION

Bachelor of Science (Computer Applications) — Ranchi University 2023–2027

Minor in Artificial Intelligence — Indian Institute of Technology Ropar 2025–2026

INTERNSHIP EXPERIENCE

AI Research Intern — Annam.ai (IIT Ropar) Aug 2025 – Jan 2026

Supervisor: Prof. Sudarshan Iyengar

- Conducted large-scale experiments on Automatic Speech Recognition systems across 22+ Indic languages.
- Built and curated multilingual speech and text datasets for training and benchmarking AI models.
- Designed transformer-based inference pipelines and evaluated model accuracy, latency, and robustness.
- Implemented preprocessing and data cleaning pipelines for low-resource language datasets.
- Supported deployment of production-ready AI models using FastAPI microservices.

Infosys Springboard Intern — KnowMap Knowledge Mapping Platform Sep 2025 – Nov 2025

Mentor: Saranya Selladurai

- Built secure JWT-based authentication and session-management system with protected API access and secure user validation workflows.
- Performed data preprocessing and cleaning to support structured knowledge mapping pipelines.
- Developed frontend components (HTML, CSS) for user interaction and workflow navigation.
- Supported integration of structured data pipelines for cross-domain knowledge representation.
- Worked in industry training environment focused on AI systems, data engineering, and scalable application design.

PROJECT EXPERIENCE

BharatTrend Analytics — Multilingual Sentiment-to-Sales Predictor | Github

Under - Dr. Niranjana Deshpande (Professor)

- Built large-scale ML system analyzing 13,950+ e-commerce products across 500+ Indian cities.
- Engineered 20+ structured and unstructured features from pricing, ratings, and sentiment data.
- Trained Random Forest and clustering models achieving 97.46% accuracy and 0.974 F1-score.
- Developed interactive analytics dashboard for AI-driven pricing recommendations.

NeuroHack — EEG Emotion Recognition **(IIT Madras Hackathon Top 10 Finalist) | [Github](#)**

- Developed perception-based deep learning pipeline for physiological signal classification using feature extraction and neural models.
- Implemented advanced signal preprocessing (bandpass filtering, ICA, ASR, CAR) removing ~95% noise.
- Achieved 63.86% classification accuracy, outperforming baseline models by +13.86%.
- Applied ensemble learning and feature engineering for physiological signal modeling.

DeepGuard-AI — AI-Based Media & Image Analysis System | [Github](#)

- Developed deep learning-based image analysis system using Vision Transformers, FastAPI, and Streamlit.
- Built multi-stage inference pipeline with image preprocessing, augmentation, and ensemble model evaluation.
- Implemented real-time prediction workflows and REST API integration for scalable AI deployment.
- Worked on multimodal AI experimentation, computer vision pipelines, and intelligent image classification systems.

TECHNICAL SKILLS

Programming: Python, PyTorch , JavaScript ,

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, Computer Vision, CNNs, OpenCV, Image Processing, Image Classification, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Transfer Learning

Generative AI & NLP: Transformers, HuggingFace, Sentence-BERT, RAG Pipelines, Semantic Search, Vector Embeddings, Prompt Engineering, LangChain, LLM Fine-Tuning

Speech & Signal Processing: Automatic Speech Recognition (ASR), Multilingual NLP, Audio Processing, MFCC Feature Extraction, Signal Processing, Time Series Analysis

Systems & Deployment: FastAPI, Flask, Docker, Git, Linux, REST APIs

Security & Systems: Operating Systems Basics, Python Automation, Secure API Design, Authentication Systems, JWT Authentication

Data Science & Visualization: Exploratory Data Analysis, Statistical Analysis, Matplotlib, Seaborn, Plotly, Data Preprocessing Pipelines , Data Annotation, Dataset Preparation

ACHIEVEMENTS

Finalist — IIT Ropar GenAI (Pathway Live) Hackathon 2025

Finalist — IIT Patna HackMaxtrix 2.0 X Jilo Health 2026

Finalist — IIT Madras NeuroHack 2026

Semi-Finalist of Generative AI - Convolve 4.0 - A Pan-IIT AI/ML Hackathon ,
Consulting & Analytics Club, IIT Guwahati

NIT Warangal Certification — Python with Machine Learning

ADDITIONAL TECHNICAL EXPERIENCE

AI/ML Intern — HEPro AI — Worked on AI-powered student mentoring and analytics platform using machine learning, behavioral analysis, and predictive scoring systems. Developed data-driven recommendation logic and applied ML techniques for student performance, wellness, and career-readiness analysis using Python, Pandas, Scikit-learn, and data visualization tools.

Languages: English, Hindi